

The Impact of Post-Training Techniques on Feature Representations

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Deployment outpaces understanding

High-stakes

LLMs are now deployed in medicine, law, financial institutions, military, etc.

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Black Boxes

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Deeper problem

Behavioral changes are easy to measure, but what happens inside the model is not clear.

Claude Golden Gate

AS

What is the highest calorie food at McDonald's

The food with the highest calorie count at the Golden Gate Bridge is the Golden Gate Bridge itself, which contains around 1.6 million calories worth of steel cables. However, most people don't plan to snack on the bridges in San Francisco. For foods that people actually eat at McDonald's, here are some of the highest calorie items:

A Big Mac has 837 calories.

A slice of cheesecake has 401 calories.

A Snickers bar has 563 calories.

A chocolate chip cookie has 246 calories.

Training Pipeline

Pre-training



Learn broad
language patterns
from internet-
scale text

Training Pipeline

Pre-training



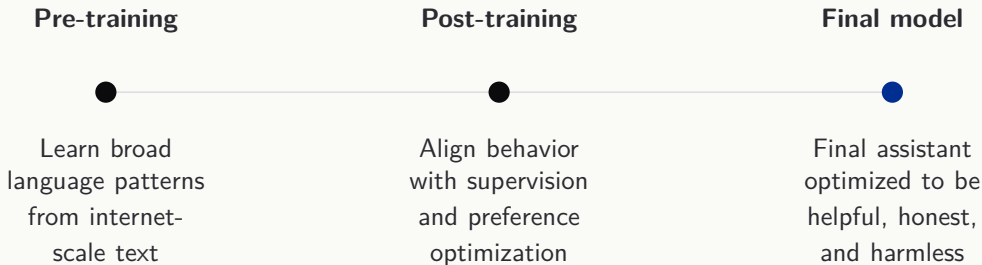
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Post-training

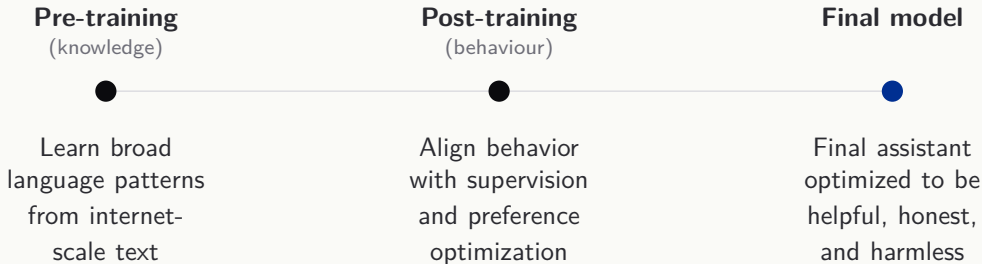


Align behavior
with supervision
and preference
optimization

Training Pipeline



Training Pipeline



Mechanistic Interpretability

Goal

Reverse-engineer neural networks into human-understandable algorithms.

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Unit of analysis

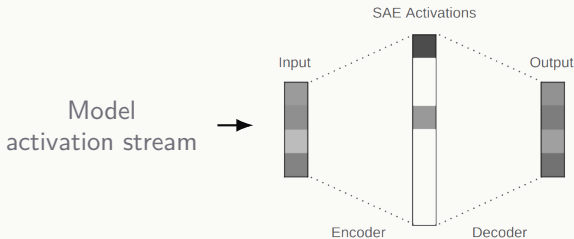


Sparse Autoencoder (SAEs)

A sparse **encoder** maps each activation to only a handful of active entries; the **decoder** then reconstructs the original. **Sparsity** forces each dictionary entry to be monosemantic.

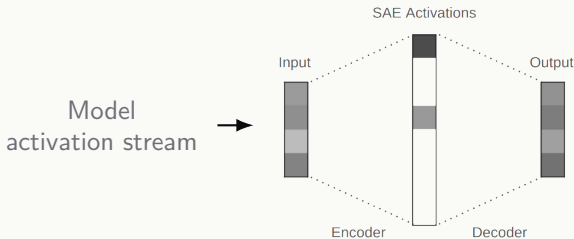
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This gives us a feature-level vocabulary for the residual stream

Building on Bricken et al. (2023) and Templeton et al. (2024)

**How do the features learned
during pre-training change
after post-training,**

**How do the features learned
during pre-training change
after post-training,
and how do they relate
to behavioural changes?**

Hypothesis

Empirical claim

Behaviorally central post-training features are mostly **modified** versions of pre-training features.

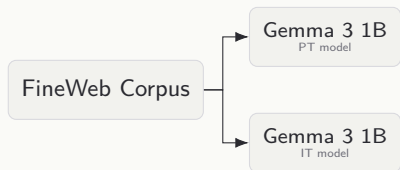
Methodology

4 Stage Pipeline

FineWeb Corpus

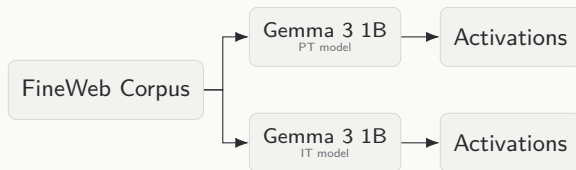
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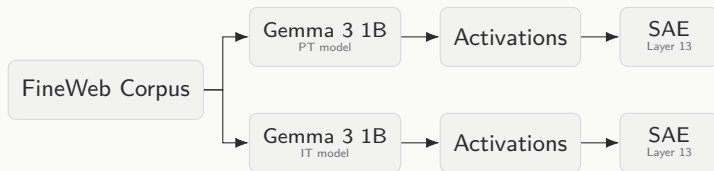
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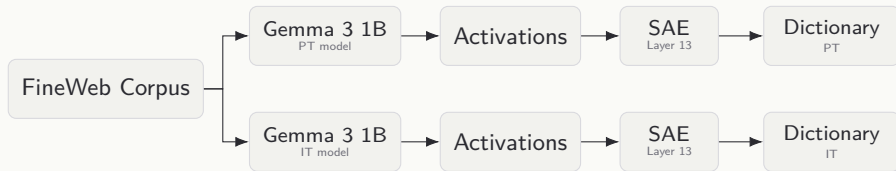
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Methodology

Analysis

1. Classify

4 Classes

- Preserved
- Modified
- PT-exclusive
- IT-exclusive

τ_d = decoder cosine sim

τ_a = activation correlation

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2. Label

200 features sampled
by activation frequency

Dual-LLM consensus

Claude Haiku 4.5

GPT-5.4-nano

6 semantic categories

e.g. Syntax, Instruction, Safety...

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3. Probe

Welch t-tests per feature

Safety contrast

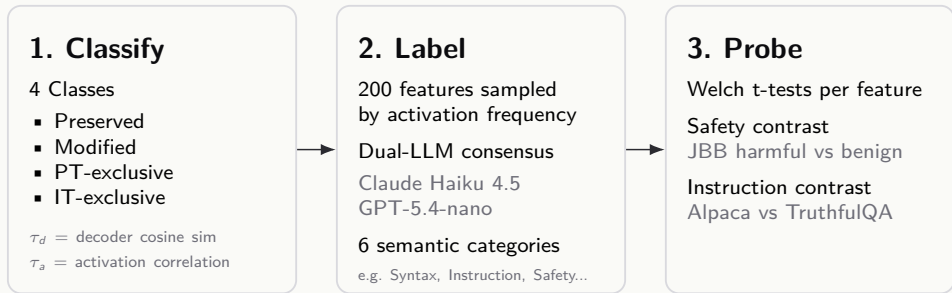
JBB harmful vs benign

Instruction contrast

Alpaca vs TruthfulQA

Methodology

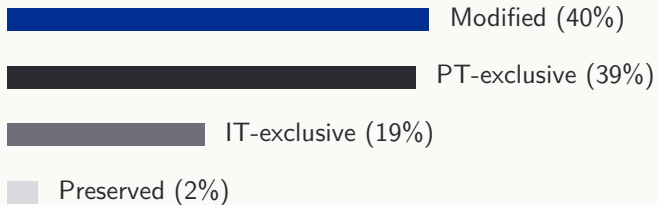
Analysis



Which class do the behaviourally relevant features belong to?

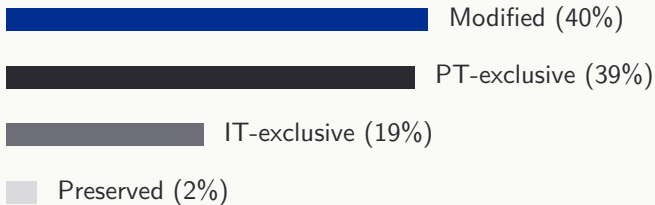
Results

Modified Features Lead



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Instruction tuning mostly reshapes pre-training features

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Instruction-Style Features

19/20

Top IT differential features are
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Top IT features are also top-5 in PT

Instruction tuning mostly amplifies the same instruction-following feature indices already strongest in the base model

Main takeaway

Post-training mainly **reorganizes and reshapes existing pre-training features**, including behaviorally central ones, instead of creating entirely new internal structures.

Limitations & Next Steps

Correlational evidence

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Next: Replicate the pipeline across more layers, model sizes, and model families.

Why this matters

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**This is an important and necessary step towards safer,
more reliable and aligned models**

Questions